```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

int main() {
    int server_fd, client_fd;
    struct sockaddr_in server_addr, client_addr;
    socklen_t client_len = sizeof(client_addr);
    char buffer[1024];

    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    if (server_fd < 0) {
        perror("Socket creation failed");
        return 1;
    }

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(8080);

    if (bind(server_fd, (struct sockaddr *)&server_addr,
            sizeof(server_addr)) < 0) {
        perror("Bind failed");
        close(server_fd);
        return 1;
    }
```

```
if (listen(server_fd, 5) < 0) {
    perror("Listen failed");
    close(server_fd);
    return 1;
}

printf("Server waiting for client...\n");

client_fd = accept(server_fd, (struct sockaddr *)&client_addr,
                   &client_len);
if (client_fd < 0) {
    perror("Accept failed");
    close(server_fd);
    return 1;
}

printf("Client connected.\n");

while (1) {
    memset(buffer, 0, sizeof(buffer));

    int n = recv(client_fd, buffer, sizeof(buffer) - 1, 0);
    if (n <= 0)
        break;

    buffer[n] = '\0';
    printf("Client: %s\n", buffer);

    if (strcmp(buffer, "exit") == 0)
        break;
```

```
        printf("Enter reply: ");
        if (fgets(buffer, sizeof(buffer), stdin) == NULL)
            break;

        buffer[strcspn(buffer, "\n")] = '\0';

        if (send(client_fd, buffer, strlen(buffer), 0) < 0)
            break;

        if (strcmp(buffer, "exit") == 0)
            break;
    }

    close(client_fd);
    close(server_fd);

    printf("Connection closed.\n");
    return 0;
}
```

Client side // code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

int main() {
   int sock;
   struct sockaddr_in server_addr;
   char buffer[1024];

   sock = socket(AF_INET, SOCK_STREAM, 0);
   if (sock < 0) {
      perror("Socket creation failed");
      return 1;
   }

   server_addr.sin_family = AF_INET;
   server_addr.sin_port = htons(8080);

   server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");

   if (connect(sock, (struct sockaddr *)&server_addr,
            sizeof(server_addr)) < 0) {
      perror("Connection failed");
      close(sock);
      return 1;
```

```
}

printf("Connected to server.\n");
printf("Type 'exit' to end the chat.\n");

while (1) {
    printf("Enter message: ");

    if (fgets(buffer, sizeof(buffer), stdin) == NULL)
        break;

    buffer[strcspn(buffer, "\n")] = '\0';

    if (send(sock, buffer, strlen(buffer), 0) < 0)
        break;

    if (strcmp(buffer, "exit") == 0)
        break;

    memset(buffer, 0, sizeof(buffer));

    int n = recv(sock, buffer, sizeof(buffer) - 1, 0);
    if (n <= 0)
        break;

    buffer[n] = '\0';
    printf("Server: %s\n", buffer);

    if (strcmp(buffer, "exit") == 0)
        break;
}
```

```
    close(sock);
    printf("Disconnected from server.\n");
    return 0;
}
```

